

Derivatives: Types & The Signals People Trade Off Them

A study companion for the Trader Screener project. Two halves: (1) the full taxonomy of derivative instruments and why each exists, and (2) the signals traders actually watch to trade them — especially the options/volatility signals that have no equivalent in plain equity screening.

Compiled 2026-06-28. Anchored to exchange/regulator and reputable-broker sources (CME, Cboe, Schwab, Barchart, QuantInsti). Commercial tools are flagged; vet for bias.

What a derivative is

A **derivative** is a contract whose value *derives* from an underlying — a stock, index, rate, currency, or commodity. You never own the underlying; you own a contract referencing it. Derivatives exist to **transfer a specific risk** (price, rate, FX, volatility, time) without trading the underlying itself, usually with **leverage** (control a large notional for a small outlay).

The recurring "why trade it" answers reduce to three: **hedge** a real exposure, take **leveraged directional** risk, or harvest **income/risk premium** (e.g., selling option premium).

PART 1 — THE TAXONOMY OF DERIVATIVES

1. Forwards

- **What:** a private, customized contract to buy/sell an asset at a set price on a future date. Traded **OTC** (over-the-counter), not on an exchange.
- **Key traits:** fully customizable (size, date, terms); **counterparty/credit risk** (no clearinghouse); no daily settlement.
- **Why trade:** bespoke hedging — a corporation locking an exact FX amount on an exact date.
- **Trade-off:** flexibility vs. illiquidity + credit risk.

2. Futures

- **What:** a **standardized, exchange-traded** forward. Fixed contract size, expiry, and terms; cleared by a central **clearinghouse**.
- **Key traits:** **margined and marked-to-market daily** (gains/losses settle each day); minimal counterparty risk; deep liquidity in major contracts.

- **Why trade:** leveraged directional exposure to commodities, rates, indices, FX, crypto; hedging real-world risk (farmer, airline, bond portfolio).
- **Sources:** CME — Introduction to Futures, CME Futures Fundamentals.

3. Options

- **What:** the **right, not the obligation**, to buy (**call**) or sell (**put**) the underlying at a set **strike** price before/at **expiration**. Buyer pays a **premium**; seller (writer) collects it and takes on the obligation.
- **Key traits:** **asymmetric payoff** (buyer's loss capped at premium, upside open; seller's gain capped at premium, risk large); value driven by strike, time, and **volatility**.
- **Why trade:** leverage, **defined-risk** hedging (protective put), and **income** (selling premium). The richest signal set of any derivative — see Part 2.
- **Sources:** CME — All About Options, Cboe Options Institute.

4. Swaps

- **What:** an agreement to **exchange streams of cash flows**. Most common: **interest-rate swaps** (fixed ↔ floating), **FX swaps**, **total-return swaps**, **credit default swaps (CDS)**.
- **Key traits:** mostly **OTC / institutional**; large notionals; used to convert the *character* of an exposure rather than bet on direction.
- **Why trade:** a firm with floating-rate debt swaps to fixed to remove rate risk; a fund gets index exposure via a total-return swap without holding the basket.
- **Note:** largely outside a retail screener's scope — included for completeness.

5. Where the lines blur (related instruments)

- **Options on futures** — an option whose underlying is a futures contract (common in commodities/rates). CME course.
- **Warrants, convertible bonds** — embedded equity options.
- **Structured products / exotics** — barrier, Asian, digital options; bespoke payoffs (Hull's text is the reference for these).

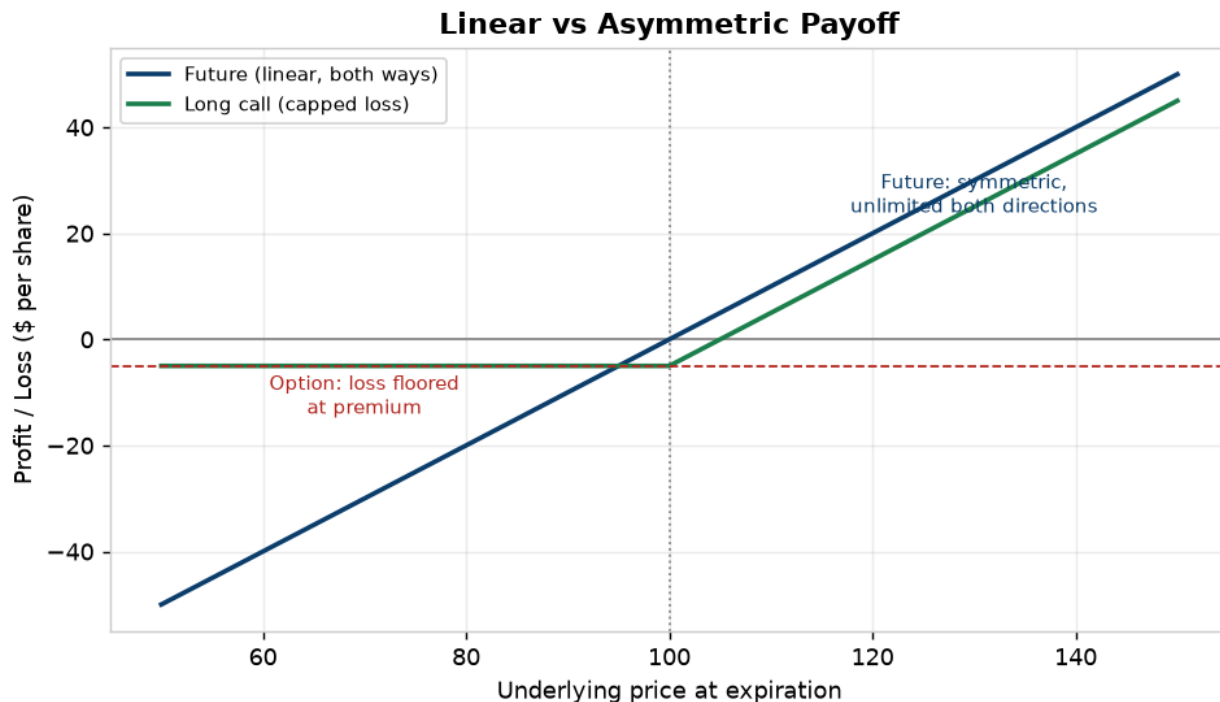
Quick comparison

Type	Venue	Standardized?	Obligation?	Main use
Forward	OTC	No	Both sides obligated	Custom hedge

Future	Exchange	Yes	Both sides obligated	Leveraged directional / hedge
Option	Exchange/OTC	Yes (listed)	Buyer optional, seller obligated	Leverage, defined-risk hedge, income
Swap	OTC (some cleared)	No	Both sides obligated	Convert exposure (rate/FX/credit)

Linear vs asymmetric payoff

A future moves dollar-for-dollar both ways (symmetric); a long option floors the loss at the premium (asymmetric). This single difference is *why* options carry the volatility signals in Part 2 — their value depends on the *probability distribution* of the move, not just its direction.

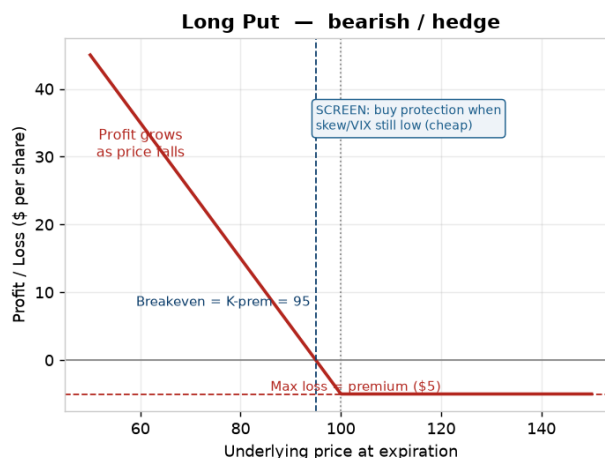
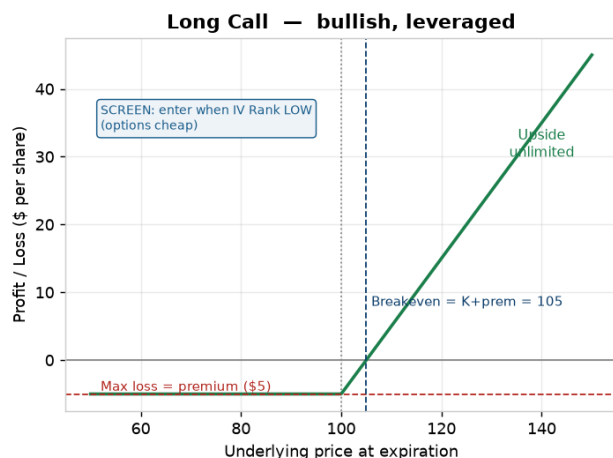


PART 1.5 — POTENTIAL ACTIONS: STRATEGY PAYOFFS & THE METRICS TO READ

Each chart shows the **profit/loss at expiration** for a common action, annotated with the three numbers you read off every payoff — **max loss**, **max profit**, **breakeven** — plus the **IV signal you'd screen for** before putting it on. (Strike $K = 100$, premium = \$5 for illustration.)

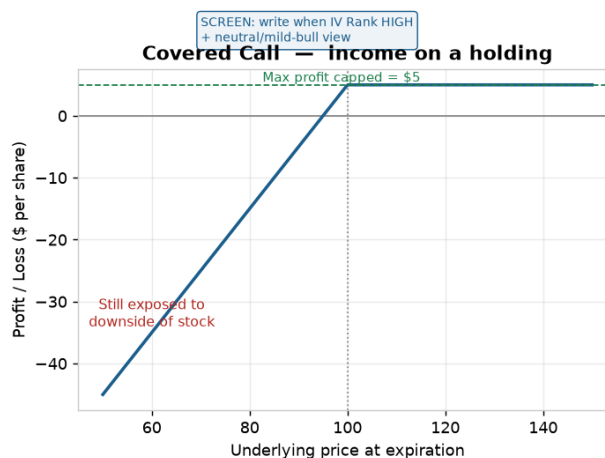
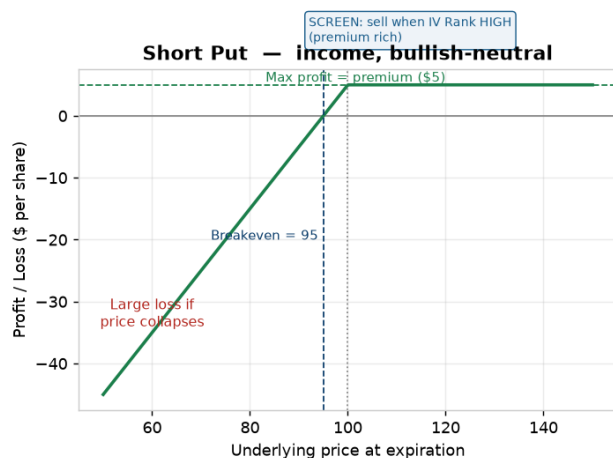
Directional / hedge — *buying* options (you pay premium, IV works against you)

- **Long call** — bullish; loss capped at premium, upside open. Best when options are **cheap** → screen for **low IV Rank**.
- **Long put** — bearish or portfolio hedge; loss capped at premium, profit grows as price falls. Buy protection *before* fear is priced in (skew/VIX still low).



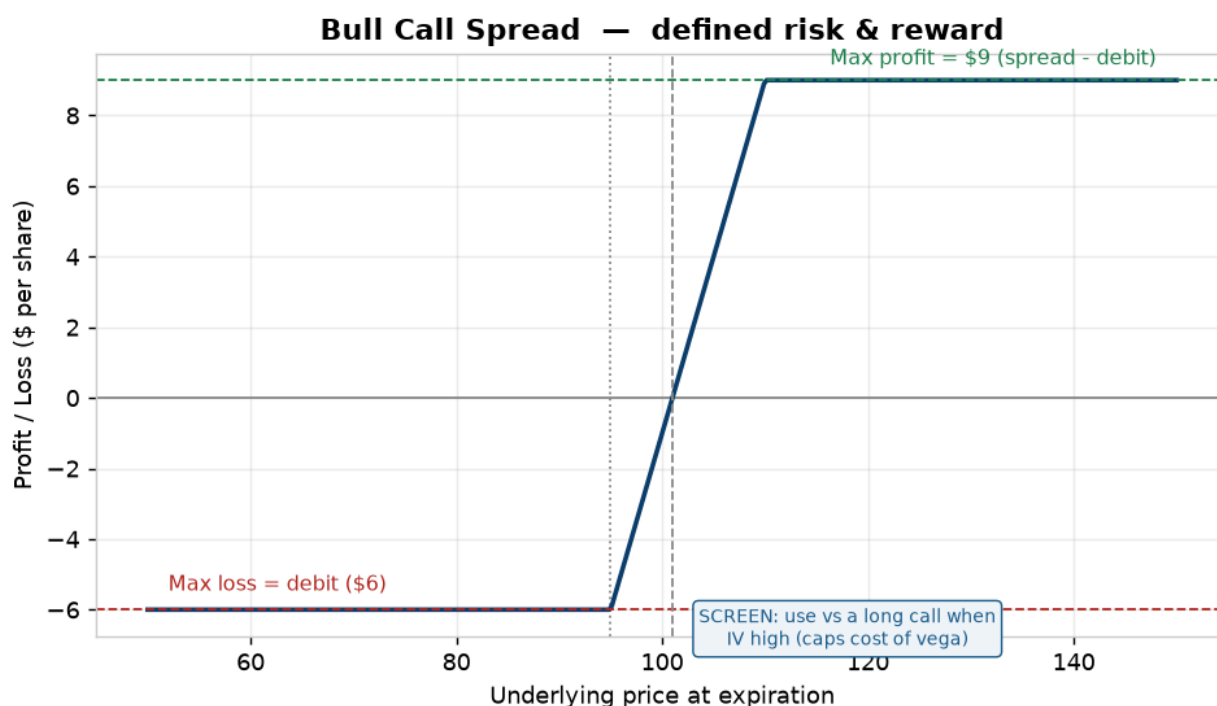
Income — *selling* options (you collect premium, IV works for you)

- **Short put** — bullish-neutral income; max profit = premium, but large loss if price collapses. Sell when premium is **rich** → screen for **high IV Rank**.
- **Covered call** — income on stock you own; caps upside at the strike in exchange for premium. Write when **IV Rank high** + a neutral/mild-bull view.



Defined risk/reward — *spreads* (cap both sides, reduce vega)

- **Bull call spread** — buy a lower strike, sell a higher one. Both max loss (the debit) and max profit (strike width – debit) are fixed. Preferred over a lone long call when **IV is high**, because selling the upper strike offsets the rich premium.



***The pattern:** buy options when IV Rank is **low** (cheap), sell/spread when IV Rank is **high** (rich). That single rule — read off the IV-Rank signal in Part 2 — drives which of these actions fits.*

PART 2 — THE SIGNALS PEOPLE TRADE OFF

DERIVATIVES

Equity screening watches price, volume, fundamentals. **Derivatives — options especially — add a second dimension: volatility and positioning.** These signals have *no equivalent* in a plain stock screener, and they're the reason options data is a different (paid) feed.

A. Implied Volatility (IV)

- **What:** the market's expectation of *future* movement, backed out of option prices. Higher IV → options more expensive → bigger move priced in.
- **Signal:** IV typically **rises into known events** (earnings, FDA, SEC filings, Fed) and **collapses after** ("IV crush").

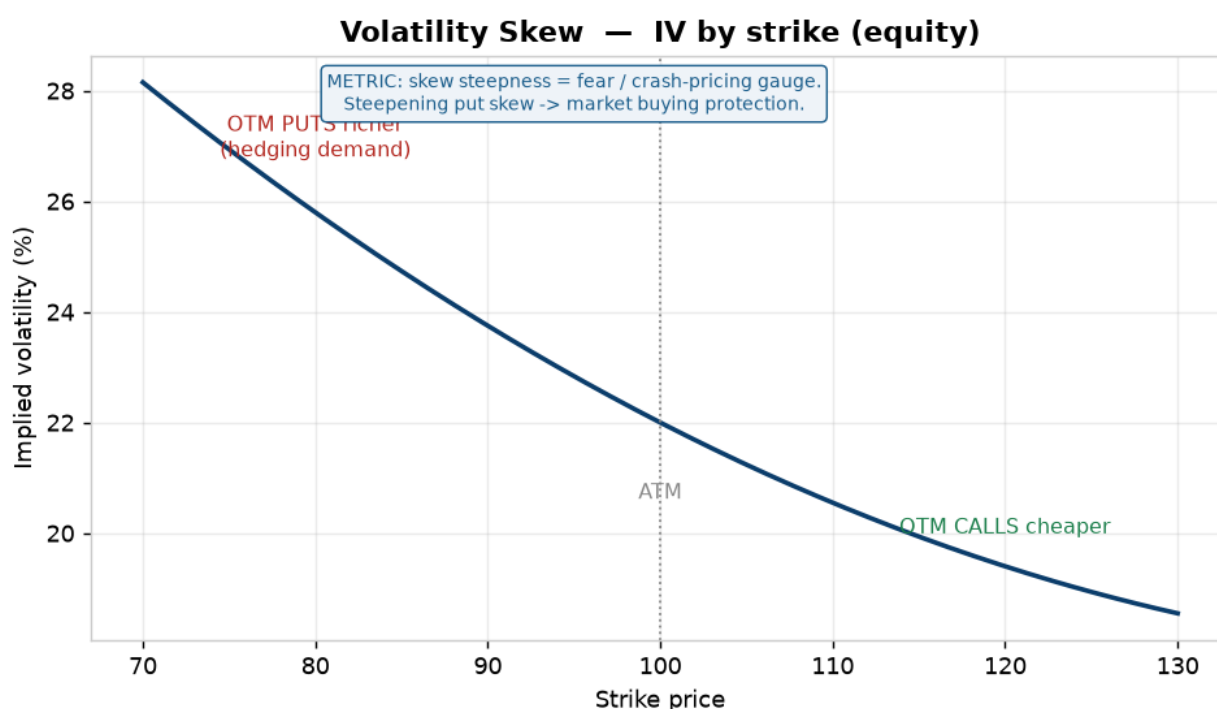
- **How traded:** **sell** premium when IV is rich, **buy** when cheap — but always relative to the name's own history (see IV Rank).
- **Sources:** Schwab — Using IV percentiles, AvaTrade — IV in options trading.

B. IV Rank & IV Percentile

- **What:** normalizers that place *current* IV against its own **trailing 1-year** range.
- **IV Rank** = position between the year's IV low and high (0–100).
- **IV Percentile** = % of days in the past year IV was *lower* than today.
- **Why it matters:** raw IV is meaningless cross-name (a biotech's "normal" IV dwarfs a utility's). Rank/percentile answer "**is IV high for this stock?**" — the actual tradeable question.
- **Sources:** Barchart — IV Rank/Percentile tool, MarketChameleon — Volatility Rankings.

C. Volatility Skew

- **What:** IV differs across **strikes** at the same expiry. Typically **OTM puts carry higher IV than OTM calls** — persistent demand for downside protection.
- **Signal:** the **shape/steepness** of skew measures fear and crash-pricing. Steepening put skew → rising demand for protection → market positioning defensively.
- **How traded:** skew informs which strikes to buy/sell (e.g., sell the rich OTM put, structure risk-reversals).
- **Sources:** QuantInsti — Trading volatility skew, Strike — Volatility skew overview.



D. Put/Call Ratio

- **What:** ratio of put to call activity (by volume or open interest), market-wide or per-name.
- **Signal:** a **sentiment gauge**. High = bearish/fearful, low = bullish/complacent. Often read **contrarian** at extremes (everyone hedged → little selling left).

E. Open Interest (OI)

- **What:** total number of **outstanding** (not-yet-closed) contracts at a strike/expiry.
- **Signal:** large OI strikes act as **support/resistance magnets** (dealers hedging around them). Rising OI + rising price = conviction behind a move; falling OI = positions closing.

F. Unusual Options Activity (UOA)

- **What:** option volume far exceeding its open interest, or large block sweeps — especially in OTM contracts.
- **Signal:** possible **informed positioning** ahead of a catalyst. Noisy, but a real flow signal.
- **Build tie-in:** this overlaps the project's **smart-money moat** — UOA is informed-flow detection in the options market, the same thesis as congress/insider tracking in equities.

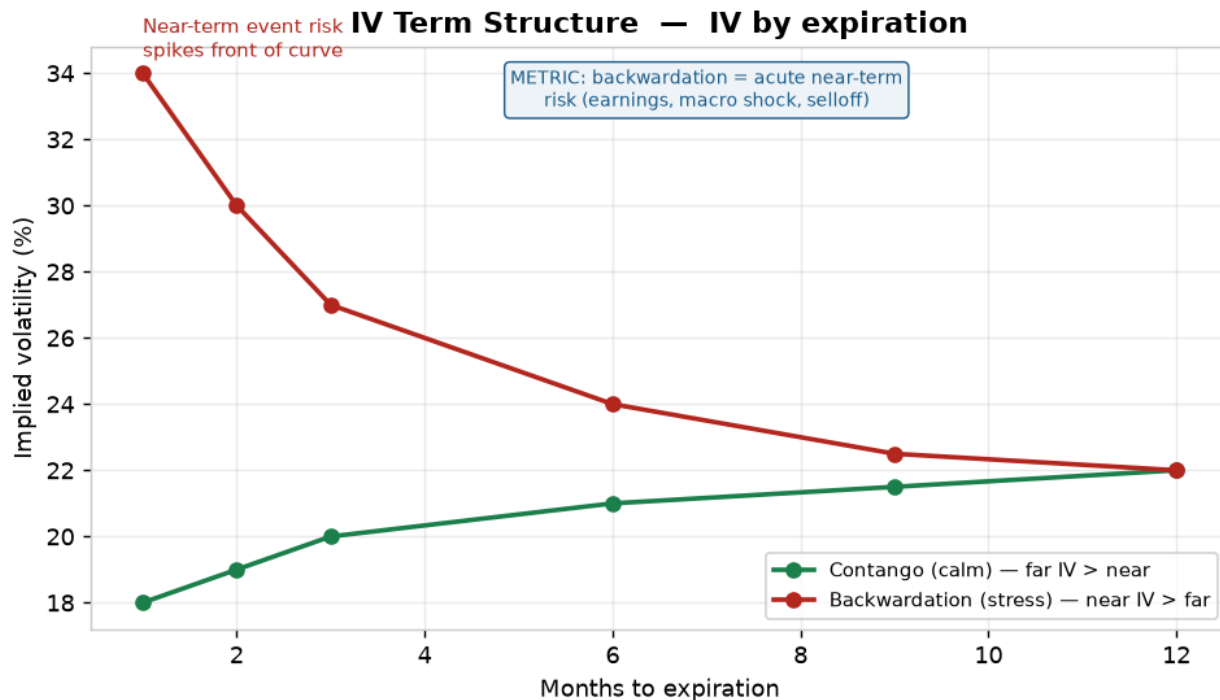
G. The Greeks (the risk dashboard)

Greek	Measures	Trader use
Delta	sensitivity to underlying price	directional exposure / hedge ratio
Gamma	rate of change of delta	acceleration risk; dealer-hedging flows (GEX)
Theta	time decay per day	premium sellers' tailwind, buyers' drag
Vega	sensitivity to IV change	exposure to volatility itself
Rho	sensitivity to interest rates	minor for short-dated

H. Term Structure of Volatility

- **What:** IV plotted across **expirations** (near vs far).

- **Signal: contango** (far IV > near) is normal/calm; **backwardation** (near IV > far) signals acute near-term event risk or stress (e.g., VIX term structure inverting in a selloff).



I. VIX & macro vol gauges

- **What:** Cboe's VIX = 30-day expected S&P volatility from option prices ("fear gauge").
- **Signal:** regime filter — spikes mark stress/capitulation; low, stable VIX marks complacency. Pairs with the project's FRED macro tab.

WHAT THIS MEANS FOR THE TRADER SCREENER

- **New data dimension, new feed.** IV, skew, OI, Greeks, UOA require an **options-chain feed** (■ paid — e.g., Polygon, ORATS, CBOE DataShop, Tradier). Free yfinance won't cover them reliably. **Decide instrument coverage (shortlist §1) before building** — options support is a real cost commitment.
- **IV Rank/Percentile is the must-have normalizer** if you screen options at all — raw IV doesn't compare across names.
- **UOA = options-side smart money.** It's the same informed-flow thesis as the congress/insider moat; if options ship, surface UOA as a flagship signal.
- **Skew + term structure + VIX = regime context**, shared with the Research and macro tabs (collect once, serve both — per [ARCHITECTURE.md](#)).

- **Signals** → **shortlist §2**. Add an options block to [SCREENER_SHORTLIST.md](#): IV, IV Rank, skew, put/call, OI, UOA, the Greeks — all ■.

Sources

- [CME — All About Options](#)
- [CME — Introduction to Futures](#)
- [CME — Options on Futures for Equity Traders](#)
- [Cboe Options Institute](#)
- [Schwab — Using Implied Volatility Percentiles](#)
- [Barchart — IV Rank & Percentile](#)
- [MarketChameleon — Volatility Rankings](#)
- [QuantInsti — How to Trade Options Using Volatility Skew](#)
- [Strike — Volatility Skew](#)
- [AvaTrade — Implied Volatility in Options Trading](#)

** Barchart, MarketChameleon, Strike, AvaTrade are commercial — useful and current; the exchange/Schwab/QuantInsti links are the durable anchors.*